The documentation and process conversion measures necessary to comply with this revision shall be completed by 5 April 2002.

INCH-POUND

MIL-PRF-19500/276D
5 January 2002
SUPERSEDING
MIL-PRF-19500/276C
1 June 2001

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, THYRISTORS (CONTROLLED RECTIFIERS), SILICON TYPES 2N2323, 2N2324, 2N2326, 2N2328, 2N2329 AND S VERSIONS 2N2323A, 2N2324A, 2N2326A, 2N2328A, 2N2329A AND S VERSIONS JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for PNPN, silicon, reverse-blocking-triode thyristors. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

* 1.2 Physical dimensions. See figure 1 (TO-5).

* 1.3 Maximum ratings.

Type	V_{RM}	V_{RM} (Non-rep)	V_{FBXM}	I_O (1)	I_{FSM} (2)	V_{GKM}	T_{OP}	T_J and T_{STG}	Barometric pressure
	V (pk)	V (pk)	V (pk)	A dc	a	V (pk)	°C	°C	mmHg
2N2323, S	50	75	50 (3)	.22	15	6	-65 to +125	-65 to +150	-
2N2323, A,S	50	75	50 (4)	.22	15	6	-65 to +125	-65 to +150	-
2N2324, S	100	150	100 (3)	.22	15	6	-65 to +125	-65 to +150	-
2N2324, A,S	100	150	100 (4)	.22	15	6	-65 to +125	-65 to +150	-
2N2326, S	200	300	200 (3)	.22	15	6	-65 to +125	-65 to +150	-
2N2326, A,S	200	300	200 (4)	.22	15	6	-65 to +125	-65 to +150	-
2N2328, S	300	400	300 (3)	.22	15	6	-65 to +125	-65 to +150	15
2N2328, A,S	300	400	300 (4)	.22	15	6	-65 to +125	-65 to +150	15
2N2329, S	400	500	400 (3)	.22	15	6	-65 to +125	-65 to +150	15
2N2329, A,S	400	500	400 (4)	.22	15	6	-65 to +125	-65 to +150	15

See notes on next page.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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1.3 Maximum ratings - Continued.

(1) This average forward current is for an ambient temperature of 80°C and 180 electrical degrees of conduction. For other operating conditions see figure 2.
(2) Surge current is non-recurrent. The rate of rise of peak surge current shall not exceed 40 A during the first 5 µs after switching from the "off" (blocking) to the "on" (conducting) state. This is measured from the point where the thyrister voltage has decayed to 90 percent of its initial blocking value.
(3) Gate connected to cathode through 1,000 ohm resistor.
(4) Gate connected to cathode through 2,000 ohm resistor.

* 1.4 Primary electrical characteristics. T_C = +25°C, unless otherwise specified.

	V_{FM}	I_{HOX} (1) (2)	V_{GT} (1)	I_{GT} (1)	V_{GT} (2)	I_{GT} (2)
	I_{FM} = 4 (pk) t_p = 8.5 ms max		2N2323 through 2N2329 2N2323S through 2N2329S T_A = -65°C thru +125°C		2N2323 through 2N2329 2N2323S through 2N2329AS T_A = -65°C thru +125°C	
	V	mA dc	V dc	µA dc	V dc	µA dc
Min			0.1		0.1	
Max	2.2	2.0	1.0	350	0.8	75

(1) Gate connected to cathode through 1,000 ohm resistor (for non-A types).
(2) Gate connected to cathode through 2,000 ohm resistor (for A types).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

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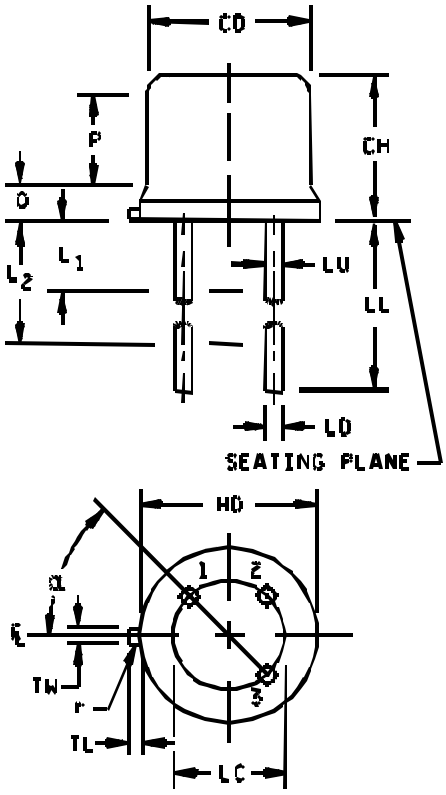


FIGURE 1. Physical dimensions.

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Ltr	Dimensions				Notes	Ltr	Dimensions				Notes
	Inches		Millimeters				Inches		Millimeters		
	Min	Max	Min	Max			Min	Max	Min	Max	
CD	.305	.335	7.75	8.51		P	.100		2.54		5
CH	.240	.2.60	6.10	6.60		r		.010		0.26	
Q					6	TL	.029	.045	0.74	1.14	9
HD	.335	.370	8.51	9.40		TW	.028	.034	0.71	0.86	
LD	.016	.021	0.41	0.53	3, 10	α	46 TP		46 TP		
LL	See notes										
LU	.016	.019	0.41	0.48	4, 10						
LC	.1414 Nom		3.592 Nom		7						

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Measured in the zone beyond .250 (6.35 mm) from the seating plane.
4. Measured in the zone .050 (1.27 mm) and .250 (6.35 mm) from the seating plane.
5. Variations on dimension CD in this zone shall not exceed .010 (0.25 mm).
6. Outline in this zone is not controlled.
7. When measured in a gauging plane .054 + .001 (1.37 + 0.03, -0.00 mm) below the seating plane of the thyristor, maximum diameter leads shall be within .007 (0.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance.
8. The anode shall be internally connected to the case.
9. Measured from the maximum diameter of the actual device.
10. All three leads.
11. For thyristor types 2N2323S, 2N2324S, 2N2326S, 2N2328S, 2N2329S, 2N2323AS, 2N2324AS, 2N2326AS, 2N2328AS LL is .500 (12.70 mm) minimum, and .750 (19.95 mm) maximum.
12. For thyristor types 2N2323, 2N2324, 2N2326, 2N2328, 2N2329, 2N2323A, 2N2324A, 2N2326A, 2N2328A LL is 1.500 (38.10 mm) minimum and 1.750 (44.45 mm) maximum
13. Pin 1 is the cathode, pin 2 is the gate, and pin 3 is the anode.

FIGURE 1. Physical dimensions - Continued.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Services (DAPS), Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-19500.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows:

I_{FBX}	DC forward blocking current with specified conditions applied to the gate.
I_{HOX}	DC holding current with specified conditions applied to the gate.
I_{RBX}	DC reverse blocking current with specified conditions applied to the gate.
R_{GK}	External resistance between gate and cathode terminals.
V_{AA}	Anode supply voltage (dc).
V_{FBX}	Forward blocking voltage, gate connected to the specified reference terminal.
V_{GG}	Gate supply voltage.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 herein.

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3.4.1 Lead finish. Lead finish shall be gold plated, tin plated or solder dipped and be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Lead material. For material compatibility, lead material shall be Kovar or Alloy 52. Where a choice of lead material is desired, it shall be specified in the acquisition document (see 6.2).

3.4.3 Construction. These devices shall be constructed in a manner and using materials which enable the thyristors to meet the applicable requirements of MIL-PRF-19500 and this document.

3.5 Marking. Devices shall be marked as specified in MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

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4.3 Screening (JANS, JANTX and JANTXV levels). Screening shall be in accordance with appendix E, table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
2	Optional	Optional
3a 3b 3c	Required Not applicable Not applicable	Required Not applicable Not applicable
4	Required	Optional
5	Required	Not applicable
7a and 7b	Required	Required
8	Required	Not required
9	I_{RBX1}, I_{FBX1}, V_{GT1}, I_{GT1}	Not applicable
10	Not applicable	Not applicable
11	I_{RBX1}, I_{FBX1}, V_{GT1}, I_{GT1}	I_{RBX1}, I_{FBX1}, V_{GT1}, I_{GT1}
12	See 4.3.2	See 4.3.1
(1) 13	Subgroups 2 and 3 of table I herein: ΔI_{RBX1} = ± 4.0 µA dc; ΔI_{FBX1} = +4.0 µA dc.	Subgroup 2 of table I herein: ΔI_{RBX1} = ± 4.0 µA dc; ΔI_{FBX1} = +4.0 µA dc.
14a and 14b	Optional	Optional
15	Required	Not required
16	Required	Not required

(1) Devices which exceed the group A limits for this test shall be rejected.

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4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_A = 125°C, t = 96 hours (min.)		
V_{FBX}	R_{GK} = 1 k ohm	R_{GK} = 2 k ohm
50 V dc	2N2323, S	2N2323A, AS
100 V dc	2N2324, S	2N2324A, AS
200 V dc	2N2326, S	2N2326A, AS
300 V dc	2N2328, S	2N2328A, AS
400 V dc	2N2329, S	2N2329A, AS

NOTE: All devices which turn “on” during the burn-in test will be removed from the lot and the quantity removed will be noted on the lot record.

4.3.2 Power burn-in for JANS. Power burn-in for JANS are as follows:

a. t = 240 hours (min).

b. T_A = 75°C, +5°C; I_O = 220 mA dc; see figure 2.

c. R_{GK} = 1 k ohm for 2N2323 through 2N2329 and 2N2323S through 2N2329S.

d. R_{GK} = 2 k ohm for 2N2323A through 2N2329A and 2N2323AS through 2N2329AS.

* 4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table V of MIL-PRF-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the test and conditions specified for subgroup testing in appendix E, table VIa (JANS) and table (JAN, JANTX and JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

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4.4.2.1 Group B inspection, appendix E, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Conditions
B4	1037	T_A = 75°C ±5°C, I_O = 220 mA dc, see figure 2. No heatsink or forced air on the device shall be permitted. R_{GK} = 1 k ohm for 2N2323 through 2N2329 and 2N2323S through 2N2329S. R_{GK} = 2 k ohm for 2N2323A through 2N2329A and 2N2323AS through 2N2329AS.
B5	1027	T_A = 175°C 96 hours or adjust to give an average lot T_J = 225°C; I_O = 220 mA dc, see figure 2.
B6	3181	$R_{\theta JA}$ = 175° C/W max.

* 4.4.2.2 Group B inspection, (JANTX and JANTXV).

Step	Method	Conditions
1.		Forward blocking voltage. See figure 3, T_A = 122°C ± 3°C. t = 170 ±24 hours. R_{GK} = 1 k ohm for 2N2323 through 2N2329 and 2N2323S through 2N2329S. R_{GK} = 2 k ohm for 2N2323A through 2N2329A and 2N2323AS through 2N2329AS. V_{FBX} =50 V dc for 2N2323, S, A, AS. V_{FBX} =100 V dc for 2N2324, S, A, AS. V_{FBX} =200 V dc for 2N2326, S, A, AS. V_{FBX} =300 V dc for 2N2328, S, A, AS. V_{FBX} =400 V dc for 2N2329, S, A, AS. 45 devices, c = 0.
2.		The forward blocking voltage test of step 1 shall be extended to 1,000 hours. Group B step 2 shall not be required more than once for any single wafer lot. 45 devices, c = 0.
3.	3181	$R_{\theta JA}$ = 175° C/W max., (for qualification only). 22 devices, c = 0.

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* 4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Must be chosen from an inspection lot that has been submitted to and passed group A, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

c. Separate samples may be used for each step. In the event of a group B failure, the manufacturer may pull a new sample at double size from either the failed assembly lot or from another assembly lot from the same wafer lot. If the new “assembly lot” option is exercised, the failed assembly lot shall be scrapped.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VII of MIL-PRF-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

* 4.4.3.1 Group C inspection, appendix E, table VII of MIL-PRF-19500.

Subgroup	Method	Conditions
C6	1037	See figure 4, T_A = 75°C ± 5°C. t_{ON} = 50 ±5 minutes, t_{OFF} =10 ±2 minutes. I_O = 220 mA dc, 1,000 cycles. R_{GK} = 1 k ohm for 2N2323 through 2N2329 and 2N2323S through 2N2329S. R_{GK} = 2 k ohm for 2N2323A through 2N2329A and 2N2323AS through 2N2329AS.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical examination 3/	2071	n = 45 devices, c = 0				
Solderability 3/ 4/	2026	n = 15 leads, c = 0				
Resistance to solvents 3/ 4/ 5/	1022	n = 15 devices, c = 0				
Temp cycling 3/ 4/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Hermetic seal 4/ Fine leak Gross leak	1071	n = 22 devices, c = 0				
Electrical measurements 4/		Group A, subgroup 2				
Bond strength 3/ 4/	2037	Precondition T_A = +250°C at t = 24 hrs or T_A = +300°C at t = 2 hrs n = 11 wires, c = 0				
Subgroup 2						
Reverse blocking current	4211	DC method, condition B	I_{RBX1}		10	µA dc
2N2323A thru 2N2329A 2N2323AS thru 2N2329AS		R_2 = 2 k ohm				
2N2323 thru 2N2329 2N2323S thru 2N2329S		R_2 = 1 k ohm				
2N2323, S, A, AS		V_R = 50 V dc				
2N2324, S, A, AS		V_R = 100 V dc				
2N2326, S, A, AS		V_R = 200 V dc				
2N2328, S, A, AS		V_R = 300 V dc				
2N2329, S, A, AS		V_R = 400 V dc				

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward blocking current	4206	DC method, condition B	I_{FBX1}		10	µA dc
2N2323 thru 2N2329 2N2323S thru 2N2329S		R_2 = 1 k ohm				
2N2323A thru 2N2329A 2N2323AS thru 2N2329AS		R_2 = 2 k ohm				
2N2323, S, A, AS 2N2324, S, A, AS 2N2326, S, A, AS 2N2328, S, A, AS 2N2329, S, A, AS		V_R = 50 V dc V_R = 100 V dc V_R = 200 V dc V_R = 300 V dc V_R = 400 V dc				
Reverse gate current	4219	V_{KG} = 6 V dc	I_{KG}		200	µA dc
Gate trigger voltage and current	4221	$V_2 = V_{FBX}$ = 6 V dc; R_L = 100 ohms				
2N2323 thru 2N2329 2N2323S thru 2N2329S		R_e = 1 k ohm	V_{GT1} I_{GT1}	0.35	0.80 200	V dc µA dc
2N2323A thru 2N2329A 2N2323AS thru 2N2329AS		R_e = 2 k ohm	V_{GT1} I_{GT1}	0.35	0.60 20	V dc µA dc
Subgroup 3						
High temperature operation:		T_A = +125°C				
Reverse blocking current	4211	DC method, condition B	I_{RBX2}		100	µA dc
2N2323A thru 2N2329A 2N2323AS thru 2N2329AS		R_2 = 2 k ohm				

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued.						
2N2323 thru 2N2329 2N2323S thru 2N2329S		R_2 = 1 k ohm				
2N2323, S, A, AS		V_R = 50 V dc				
2N2324, S, A, AS		V_R = 100 V dc				
2N2326, S, A, AS		V_R = 200 V dc				
2N2328, S, A, AS		V_R = 300 V dc				
2N2329, S, A, AS		V_R = 400 V dc				
Forward blocking current	4206	DC method, condition B	I_{FBX2}		100	µA dc
2N2323 thru 2N2329 2N2323S thru 2N2329S		R_2 = 1 k ohm				
2N2323A thru 2N2329A 2N2323AS thru 2N2329AS		R_2 = 2 k ohm				
2N2323, S, A, AS		V_{FBX} = 50 V dc				
2N2324, S, A, AS		V_{FBX} = 100 V dc				
2N2326, S, A, AS		V_{FBX} = 200 V dc				
2N2328, S, A, AS		V_{FBX} = 300 V dc				
2N2329, S, A, AS		V_{FBX} = 400 V dc				
Gate trigger voltage	4221		V_{GT3}	0.1		V dc
2N2323, S, A, AS 2N2324, S, A, AS 2N2326, S, A, AS		R_L = 1,000 ohm; R_e = 1 k ohm				
2N2328, S, A, AS 2N2329, S		R_L = 100 ohm R_e = 2 k ohm				
2N2323, S, A, AS		$V_2 = V_{FBX}$ = 6 V dc				
2N2324, S, A, AS		$V_2 = V_{FBX}$ = 6 V dc				
2N2326, S, A, AS		$V_2 = V_{FBX}$ = 6 V dc				
2N2328, S, A, AS		$V_2 = V_{FBX}$ = 6 V dc				
2N2329, S, A, AS		$V_2 = V_{FBX}$ = 6 V dc				

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued.						
Low temperature operation		T_A = -65°C				
Reverse blocking current	4211	DC method, condition B	I_{RBX3}		5	µA dc
2N2323A thru 2N2329A 2N2323AS thru 2N2329AS		R_2 = 2 k ohm				
2N2323 thru 2N2329 2N2323S thru 2N2329S		R_2 = 1 k ohm				
2N2323, S, A, AS 2N2324, S, A, AS 2N2326, S, A, AS 2N2328, S, A, AS 2N2329, S, A, AS		V_R = 50 V dc V_R = 100 V dc V_R = 200 V dc V_R = 300 V dc V_R = 400 V dc				
Forward blocking current	4206	DC method, condition B	I_{FBX3}		5	µA dc
2N2323 thru 2N2329 2N2323S thru 2N2329S		R_2 = 1 k ohm.				
2N2323A thru 2N2329A 2N2323AS thru 2N2329AS		R_2 = 2 k ohm.				
2N2323, S, A, AS 2N2324, S, A, AS 2N2326, S, A, AS 2N2328, S, A, AS 2N2329, S, A, AS		V_{FBX} = 50 V dc V_{FBX} = 100 V dc V_{FBX} = 200 V dc V_{FBX} = 300 V dc V_{FBX} = 400 V dc				
Gate trigger voltage and current	4221	$V_2 = V_{FBX}$ = 6 V dc; R_L = 100 ohms				
2N2323 thru 2N2329 2N2323S thru 2N2329S		R_e = 1 k ohm	V_{GT2} I_{GT2}		1.0 350	V dc µA dc
2N2323A thru 2N2329A 2N2323AS thru 2N2329AS		R_e = 2 k ohm	V_{GT2} I_{GT2}		0.8 75	V dc µA dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Exponential rate of voltage rise	4231	T_A = 125°C, condition B, 50 ohms ≤ R_L ≤ 400 ohms, C = 0.1 to 1.0 µF, repitition rate = 60 pps, test duration = 15 seconds	V_{FBX}			
2N2323 thru 2N2329 and 2N2323S thru 2N2329S		dv/dt = 1.8 V/µs, R_3 = 1 k ohm				
2N2323A thru 2N2329A and 2N2323AS thru 2N2329AS		dv/dt = 0.7 V/µs, R_3 = 2 k ohm				
2N2323, S, A, AS		V_{AA} = 50 V dc		47		V dc
2N2324, S, A, AS		V_{AA} = 100 V dc		95		V dc
2N2326, S, A, AS		V_{AA} = 200 V dc		190		V dc
2N2328, S, A, AS		V_{AA} = 300 V dc		285		V dc
2N2329, S, A, AS		V_{AA} = 400 V dc		380		V dc
Forward “on” voltage	4226	I_{FM} = 4 A(pk) (pulse), pulse width = 8.5 ms, max; duty cycle = 2 percent max	V_{FM}		2.2	V (pk)
Holding current	4201	Condition B, V_{AA} = 24 V dc max, I_{F1} = 100 mA dc, I_{F2} = 10 mA dc, gate trigger source voltage = 6 V dc, trigger pulse width = 25 µs minimum, R_2 = 330 ohms	I_{HOX}		2.0	mA dc
2N2323 thru 2N2329 2N2323S thru 2N2329S		R_3 = 1 k ohm				
2N2323A thru 2N2329A 2N2323AS thru 2N2329AS		R_3 = 2 k ohm				
Subgroup 5						
Not applicable						

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 6						
Surge current	4066	See figure 4; T_A = 122°C, I_{FM} = 15 A; 10 surges, 1 per minute, I_O = 0 at rated V_{RWM}, f = 60 Hz				
End-point electrical measurements		See table II, steps 1, 2, 3 and 4				
Subgroup 7						
Nonrepetitive peak voltage		See figure 5, 10 pulses.				
2N2323, S, A, AS		V_{RM} = 75 V dc				
2N2324, S, A, AS		V_{RM} = 150 V dc				
2N2326, S, A, AS		V_{RM} = 300 V dc				
2N2328, S, A, AS		V_{RM} = 400 V dc				
2N2329, S, A, AS		V_{RM} = 500 V dc				
Electrical end-point measurements		See table II, steps 1, 2, 3,and 4.				

1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed subgroup A1, double the sample size of the failed test or sequence of tests. A failure in group A, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for JANS devices.
5/ Not required for laser marked devices.

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TABLE II. Groups A, B and C electrical end-point measurements. 1/ 2/ 3/ 4/

Step	Inspection	MIL-STD-760		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Reverse blocking current 2N2323 thru 2N2329 2N2323S thru 2N2329S 2N2323A thru 2N2329A 2N2323AS thru 2N2329AS 2N2323, S, A, AS 2N2324, S, A, AS 2N2326, S, A, AS 2N2328, S, A, AS 2N2329, S	4211	DC method, condition C R_2 = 1 k ohm R_2 = 2 k ohm V_R = 50 V dc V_R = 100 V dc V_R = 200 V dc V_R = 300 V dc V_R = 400 V dc	I_{RBX1}		10	µA dc
2.	Forward blocking current 2N2323 thru 2N2329 2N2323S thru 2N2329S 2N2323A thru 2N2329A 2N2323AS thru 2N2329AS 2N2323, S, A, AS 2N2324, S, A, AS 2N2326, S, A, AS 2N2328, S, A, AS 2N2329, S	4206	DC method, condition C R_2 = 1 k ohm R_2 = 2 k ohm V_{FBX} = 50 V dc V_{FBX} = 100 V dc V_{FBX} = 200 V dc V_{FBX} = 300 V dc V_{FBX} = 400 V dc	I_{FBX1}		10	µA dc
3.	Forward “on” voltage	4226	I_{FM} = 4A(pk) (pulse), pulse width = 8.5 ms, max; duty cycle = 2 percent max	V_{FM}		2.2	V (pk)
4.	Gate trigger voltage	4221	$V_2 = V_{FBX}$ = 6 V dc; R_L = 100 ohms	V_{FM}			
	2N2323 thru 2N2329 and 2N2323S thru 2N2329S		R_e = 1 k ohm	V_{GT1} I_{GT1}	0.35	0.8 200	V dc µA dc
	2N2323A thru 2N2329A 2N2323AS thru 2N2329AS		R_e = 2 k ohm	V_{GT1} I_{GT1}	0.35	0.6 20	V dc µA dc

See footnotes on next page.

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TABLE II. Groups A, B and C electrical end-point measurements - Continued. 1/ 2/ 3/ 4/

1/ The electrical measurements for appendix E, table V of MIL-PRF-19600 are as follows: Subgroup 7, see table II herein, steps 1, 2, 3 and 4.

2/ The electrical measurements for appendix E, table VIa (JANS) of MIL-PRF-19500 are as follows:
   a. Subgroup 3, see table II herein, steps 1, 2, 3 and 4.
   b. Subgroup 4, see table II herein, steps 1, 2, 3 and 4.
   c. Subgroup 5, see table II herein, steps 1, 2, 3 and 4.

3/ The electrical measurements for appendix E, table VIb (JANTX and JANTXV) of MIL-PRF-19600 are as follows:
   a. Step 1, see table II herein, steps 1, 2, 3 and 4.
   b. Step 2, see table II herein, steps 1, 2, 3 and 4.
   c. Step 3, see table II herein, steps 1, 2, 3 and 4."

4/ The electrical measurements for appendix E, table VII of MIL-PRF-19500 are as follows:
   a. Subgroup 2, see table II herein, steps 1, 2, 3 and 4.
   b. Subgroup 3, see table II herein, steps 1, 2, 3 and 4.
   c. Subgroup 6, see table II herein, steps 1, 2, 3 and 4.

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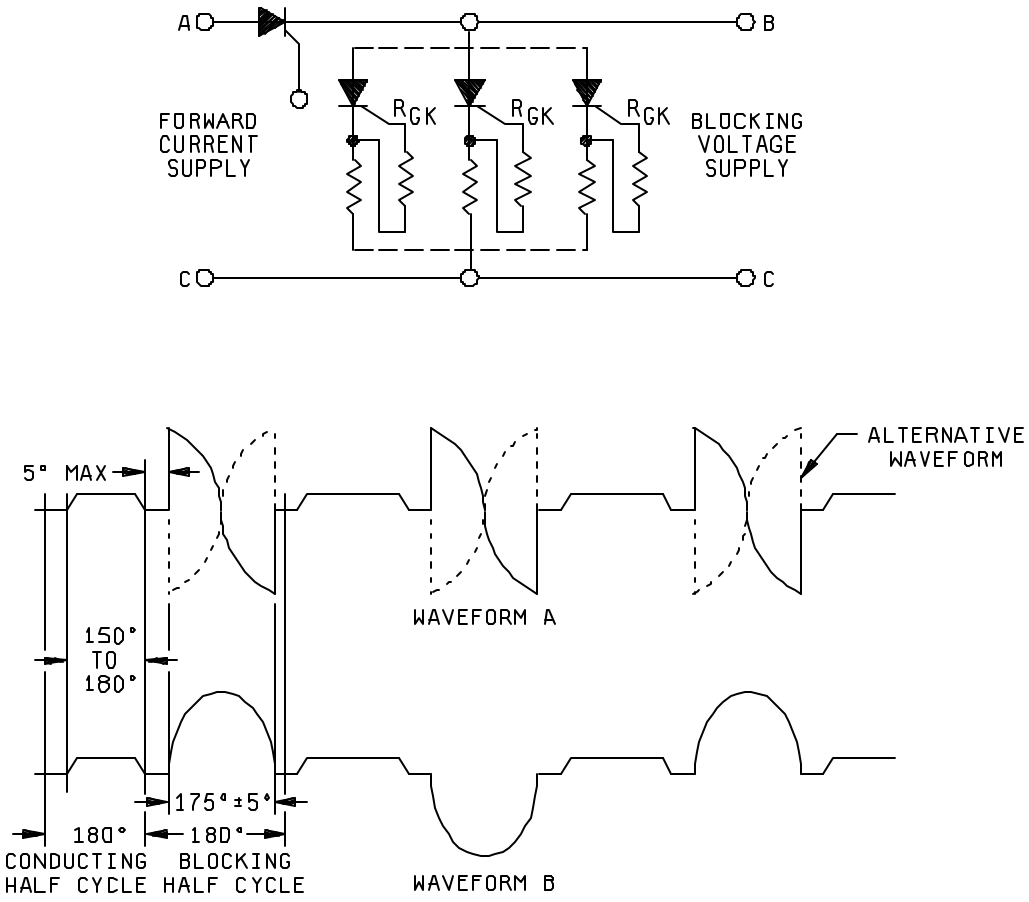


FIGURE 2. Test circuit and waveforms for intermittent operation life test.

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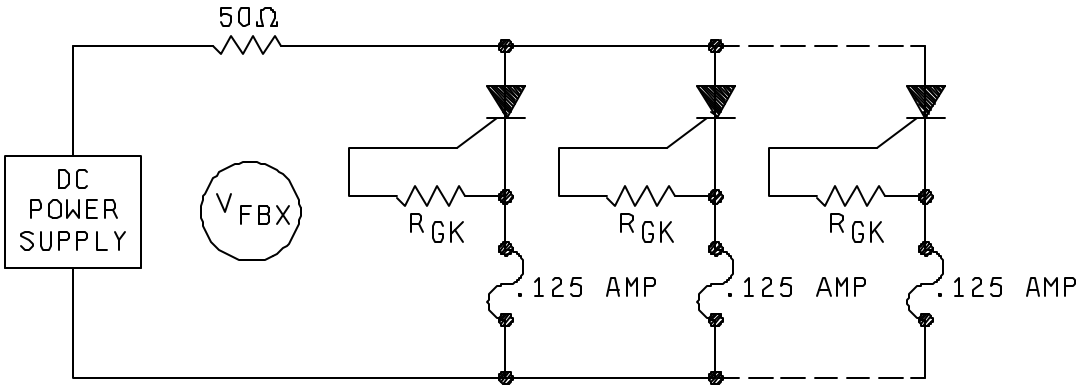
Test circuit details: The test circuit used must provide one of the waveforms shown. When a large number of devices are to be tested, they may be connected in parallel or series combinations with suitable provisions (resistive, reactive, etc, methods) for proper division of forward current or blocking voltages. Suitable protective elements should be used to isolate defective devices without interupting the test and to protect the remaining devices on test. The forward current suppy shall consist of a low-voltage transformer with a means of adjusting the secondary voltage. It shall supply 22 mA dc average forward current to each device under test. The gate supply shall be in phase with the forward current supply. The average power input shall not be allowed to exceed .01 watt per device.

Operating condtions: The devices shall be subjected to the following operational conditions:

a. Power sources are to be 60 Hz sinusoidal waveform sources.
b. The conduction angle of I_O shall be 150 degrees to 180 degrees.
c. The rated forward or reverse blocking voltage shall be started during the nonconducting half cycle no later than 5 degrees after conduction has ceased. For waveform A, the IRC time constant of the blocking voltage shall be 200 ±100 microseconds.
d. For accelered steady-state operation, blocking voltage shall be equal to or greater than 30 V.
e. The duration of the blocking voltages shall be 175 ±5 degrees.
f. The duration of life test shall be 1,000 (+72, -24) hours.
g. Ambient test temperature shall be 75 ±5°C.

FIGURE 2. Test circuit and waveforms for intermittent operation life test - Continued.

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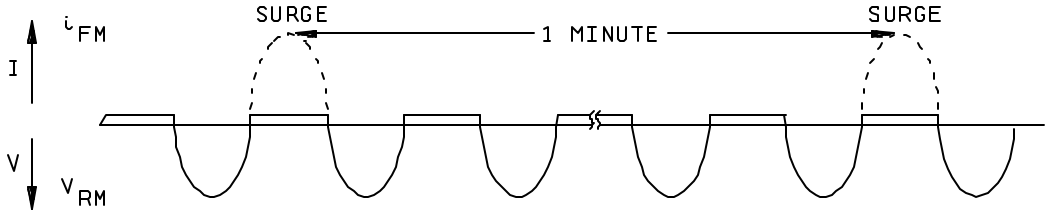


Procedure: Adjust the power supply to the specified value for the device under test.

NOTES:
1. Rate of rise of voltage must be limited to values specified in group A inspection.
2. One-eight ampere fuse may be replaced by an equivalent load protecting device.

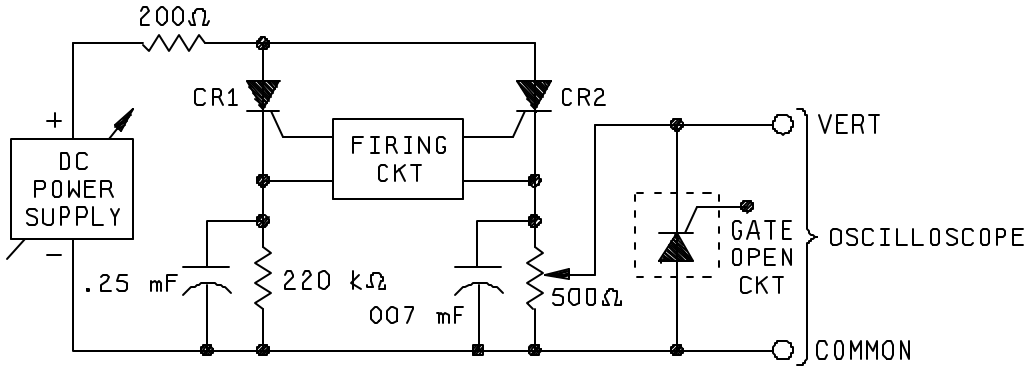
FIGURE 3. Blocking-voltage test.

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Procedure: The above waveform shall be applied to the devices under test. During this test, the gate shall be connected to the anode through a suitable firing resistor and series diode or fired by an equivalent method. The device shall be operated at the specified maximum reverse voltage and specified temperature in a single phase circuit with a 60 Hz supply and a resistive load. The total peak device current during the surge shall be 15 amperes.

FIGURE 4. Surge current waveform.



Procedure: Adjust potentiometer to achieve the specified voltage across the device under test. Timing sequence of firing circuit is: Apply trigger pulse to CR2 once every 15 seconds, apply trigger pulse to CR1, 5.0 ±1.0 milliseconds after firing CR2. The specified number of pulses shall be applied. Alternate circuits shall apply rectangular pulses of 5.0 ±1.0 milliseconds with a maximum rise and fall time of 50 microseconds.

FIGURE 5. Nonrepetitive peak reverse voltage test.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.2.1).

c. The lead finish or material as specified (see 3.4.1 and 3.4.2).

d. Type designation and quality assurance level.

e. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturer’s List whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from, Defense Supply Center Columbus, ATTN: DSCC-VQE, P.O. Box 3990, Columbus, OH 43216-5000.

6.4 Changes from previous issue. The margins of this revision are marked with an asterisk to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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Preparing activity:
DLA - CC

(Project 5961-2500)

Review activities:
Army – MI, SM
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/276D	2. DOCUMENT DATE 5 January 2002

3. **DOCUMENT TITLE**
SEMICONDUCTOR DEVICE, THYRISTORS (CONTROLLED RECTIFIERS), SILICON TYPES 2N2323, 2N2324, 2N2326, 2N2328, 2N2329 AND S VERSIONS 2N2323A, 2N2324A, 2N2326A, 2N2328A, 2N2329A AND S VERSIONS JAN, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) COMMERCIAL DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial DSN FAX EMAIL 614-692-0510 850-0510 614-692-6939 alan.barone@dscc.dla.mil
c. ADDRESS Defense Supply Center, Columbus ATTN: DSCC-VAC, P.O. Box 3990 Columbus, OH 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman, Suite 2533, Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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